

# Sishir Subedi

Data science professional with expertise in bioinformatics, deep learning, and cancer testing.

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## EDUCATION

### PhD, Bioinformatics

2022 - 25

University of British Columbia, Vancouver, Canada

- Fellowship, three first-author machine learning methods papers

### MS, Computer Science (GPA 3.75)

2016 - 18

Texas A&M University, College Station, USA

- Research assistantship, one co-author bioinformatics analysis paper
- Post-bacc, Computer Science (GPA 4.0) - University of Houston, USA

2014 - 16

### BS, Molecular Biology (GPA 3.98)

2008 - 11

Texas Tech University, Lubbock, USA

- Awards for academic and research achievements

## PROFESSIONAL EXPERIENCE

### HOUSTON METHODIST HOSPITAL, TX, USA

Bioinformatician

June 2018 - Oct 2021

- Design and implement bioinformatics pipelines for cancer assays, real-time pathology analytics, and clinical tools. Collaborate with internal cross-functional teams and third-party vendors for data management and systems integration in the CLIA-certified diagnostics laboratory.
- Sequencing (Illumina, Ion, Nanopore), pipelines (cancer panel, WES, TMB, COVID), workflow (snakemake, singularity, conda), EHR/clinical reporting (Java, SQL, Epic systems, Tableau), HPC (PBS Torque, Azure)

### NEUDESIC, TX, USA

IT Associate Intern

Dec 2014 - Apr 2016

- Deliver technology-driven solutions to Fortune 100 energy clients, custom application support for risk assessment, analytics, and quality assurance. SQL Server with ASP.NET applications.

### WASHINGTON UNIVERSITY IN ST LOUIS, MO, USA

Research Technician

Jun 2011 - May 2013

- Investigate the molecular mechanisms of learning and memory. Training in molecular biology techniques (cloning, qRT-PCR, in situ hybridization, western blotting) including mouse genetics, cell culture, and fluorescence microscopy.

## RESEARCH EXPERIENCE

### BRITISH COLUMBIA CANCER CENTER, BC, CANADA

*Graduate Research Assistant (PhD)*

*Jan 2022 - Present*

- Develop novel computational methods for large-scale omics data for interpretable deep learning models, scalable annotation, and multimodal data integration.
- Languages (Python, C++, R, Bash), deep learning (NNs, CNNs, RNNs, Transformers, GNNs, VAEs), libraries (PyTorch, scikit-learn), data science (feature engineering, latent variable models, optimization)

### TEXAS A&M UNIVERSITY, TX, USA

*Graduate Research Assistant (MS)*

*Aug 2016 - Jan 2018*

- Develop machine learning methods to analyze the lifecycle of a pathogen. Bulk RNAseq analysis pipelines.

### BAYLOR COLLEGE OF MEDICINE, TX, USA

*Research Assistant*

*June 2013 - Jul 2014*

- First year of biomedical graduate program with rotation projects in three labs (in-vitro drug screening, neuronal stem cells, and genetic model for ALS disease). Transfer to pursue MS in CS.

### TEXAS TECH HEALTH SCIENCE CENTER, TX, USA

*TTU-HHMI Research Scholar (BS)*

*May 2009 - Apr 2011*

- Study of signaling pathways during embryogenesis in drosophila. Genetic crosses and expression analysis.

## SELECTED PUBLICATIONS

- **S Subedi** and Y Park. (2025) Decomposing patient heterogeneity of single-cell cancer data by cross-attention neural networks (In prep).
- **S Subedi**, T. Sumida, and Y Park. (2024) A scalable approach to topic modelling in single-cell data by approximate pseudobulk projection. Life Science Alliance, 7(10).
- **S Subedi** and Y Park. (2023) Single-cell pair-wise relationships untangled by composite embedding model. Iscience, 26(2).
- P. Christensen, R. Olsen, S. Long, **S Subedi**, *et al.* (2022) Delta variants of SARS-CoV-2 cause significantly increased vaccine breakthrough COVID-19 cases in Houston, Texas. The American journal of pathology 192 (2), 320-331.
- P. Hodjat, P. Christensen, **S. Subedi**, *et al.* (2021) The reemergence of seasonal respiratory viruses in Houston, Texas, after relaxing COVID-19 restrictions. Microbiology Spectrum 9 (2), e00430-21.
- S. Jianming, **S. Subedi**, *et al.* (2021) Identifying possible germline variants from tumor-only sequencing of hematological malignancies. Leukemia & Lymphoma 62, no. 2: 482-485.
- G. Eskandari, **S. Subedi**, *et al.* (2021) Implementing flowDensity for automated analysis of bone marrow lymphocyte population. Journal of Pathology Informatics 12 (1), 49.

- S. Long, R. Olsen, [ et al., including **S.Subedi** ]. (2020) Molecular Architecture of Early Dissemination and Massive Second Wave of the SARS-CoV-2 Virus in a Major Metropolitan Area. *mBio*, 11(6), e02707-20.
- A. Bandekar, **S. Subedi**, T. Ioerger, C. Sasseti. (2020) Cell cycle-associated expression patterns predict gene function in mycobacteria. *Current Biology*, 30(20), 3961-3971.
- E. Salazar, K. Perez, [ et al., including **S. Subedi** ]. (2020) Treatment of Coronavirus Disease 2019 (COVID-19) Patients with Convalescent Plasma. *The American Journal of Pathology*, 190(8), 1680–1690.
- P. Christensen, **S. Subedi**, *et al.* (2020) Development and Validation of Houston Methodist Variant Viewer version 3: Updates to an application to support clinical interpretation of next-generation sequencing data for cancer. *JAMIA open*, 3(2), 299-305.
- **S. Subedi**, Y. Li, C. Early, *et al.* (2016) System for the Analysis of EEG Data and Brain State Modeling. *Emerging Trends in Computational Biology, Bioinformatics, and Systems Biology - Systems & Applications*, 2016, 447-465.
- A. Chan, C. Early, **S. Subedi**, Y. Li, and H. Lin. (2015) Systematic analysis of machine learning algorithms on EEG data for brain state intelligence. *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* (pp. 793-799).

## AWARDS

- Doctoral Fellowship (4YF), Faculty of Science - Bioinformatics, UBC, 2022
- Full Tuition Scholarship and Research Assistantship, Dept. of Computer Science, TAMU, 2016
- Outstanding Student in Computer Science, College of Sciences, UHD, 2015
- Full Tuition Scholarship Award and Research Assistantship, BCM, 2013
- Earl Camp Award for Outstanding Graduating Senior, Dept. of Biological Sciences, TTU, 2011
- Dr. Richard L. Blanton Endowed Scholarship for Undergraduate Research, HHMI-TTU Research, 2010
- Presidential Scholarship, College of Sciences, TTU, 2009